

Solutions and Solubility — Paper B

MEDIUM • NON-CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says **explain**, your answer must talk about **particles** — solute particles spreading between solvent particles, collisions, or how much the solvent can hold at that temperature.
- Solubility is always given in **g per 100 g of water**. If a question uses a different mass of water, scale it.
- The mark for each question is shown in square brackets on the right.
- Reminder: stirring, crushing and heating change how **fast** something dissolves. Only temperature changes how **much**.

SECTION 1 – SATURATED SOLUTIONS

- 1.** Describe how you could make a saturated solution of salt in water, and how you would know it was saturated. [3]

- 2.** The solubility of potassium nitrate at 40 °C is 64 g per 100 g of water.

- (a)** Calculate the mass that will dissolve in 25 g of water at 40 °C. [2]

- (b)** Calculate the mass that will dissolve in 250 g of water at 40 °C. [2]

3. Use the table.

Solubility (g per 100 g water)

Temperature (°C):	0	20	40	60	80	100
potassium nitrate:	13	32	64	110	169	246
copper sulfate:	14	21	29	40	55	75
sodium chloride:	36	36	37	37	38	40

- (a)** State the solubility of copper sulfate at 60 °C. [1]

- (b)** Which of the three substances is most soluble at 20 °C? Which is most soluble at 60 °C? [2]

- (c)** A solution contains 30 g of potassium nitrate in 100 g of water at 30 °C. Is it saturated? Explain, using a value from the table. [2]

- (d)** Which substance would you **not** bother heating to dissolve more of? Give a reason from the data. [2]

- 4.** A saturated solution of potassium nitrate in 100 g of water is made at 60 °C and cooled to 20 °C. Use the table to calculate the mass of crystals that forms. Show your working. [3]

- 5.** Explain why crystals form when the solution in the previous question is cooled. [2]

6. For each change, explain in terms of particles why it makes a solid dissolve faster.

(a) Stirring the solution.

[2]

(b) Crushing the solid into a powder.

[2]

(c) Warming the water.

[2]

7. Which **one** of the three changes in the previous question also increases how **much** can dissolve? Explain why the other two do not.

[3]

8. Explain why a fizzy drink goes flat faster when it is warm.

[2]

End of paper. Check every answer has a unit or a form the question actually asked for.